

Fig. 2: Actual-size PCB layout of the long delay timer with real-time clock display

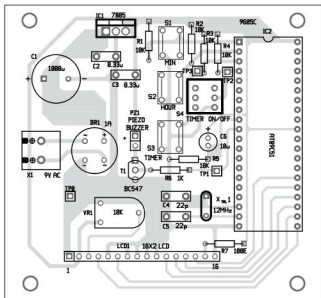


Fig. 3: Component layout of the PCB

LCD display. In this application, LCD library function is used to display the time and countdown timer on a 16x2 LCD, controlled by an HD44780 LCD controller. RS and EN control pins of LCD are connected to port pins P2.0 and P2.1 of IC2, respectively. IC2 pins 23, 24, 25 and 26 are connected to data pins (11-14) of the LCD. Pins 7, 8, 9, 10 and 16 of the LCD are connected to ground and pin 15 is connected to Vcc through a 100-ohm resistor. Since the data is read from the microcontroller, R/W pin 5 of LCD is also grounded.

The LCD displays real-time in hours : minutes : seconds format with AM or PM, and the countdown timer with on/off status. In this circuit, the LCD is programmed in 4-bit mode, so it needs only four data bits. By using switch S1, you can vary the minutes from zero to 59 and switch S2 to set the hour from one to 12. Similarly, switch S3 is used for changing the countdown timer. Initially, the countdown timer is at zero. When it is set to a value more than zero, it will start decrementing based on the status of seconds in real-time clock. That is, whenever the seconds reach zero after crossing 59, the timer decrements by one count.

Switch S4 not only controls the buzzer but also controls the countdown timer display on the LCD.

Initially, switch S4 is set to off position.

Power supply. The circuit requires a 5V supply for driving the LCD and microcontroller. Here, the power supply is derived from 230V AC primary to 9V, 500mA secondary transformer. IC 7805 converts the 9V supply to a regulated 5V DC supply.

Software program

The program for the microcontroller is written in C language and compiled using MikroC compiler. The compiler is used to convert the C program into a hex file for burning into the Flash memory of AT89C51 microcontroller using a suitable programmer. MikroC compiler is available for various microcontrollers like 8051, PIC, AVR and ARM. Here, MikroC compiler for 8051 is chosen that has a code limit of 2KB.

The program uses timer 0 interrupt as given below.

```
void Timer0InterruptHandler() org IWT_
ADDR_ET0{
```

It is programmed in mode 1 as a 16-bit timer. The interrupt enable bit and timer 0 interrupt bit are enabled initially. Then, timer 0 is programmed to give an interrupt every 50ms, accurately. Timer run bit TR0 is set to one to run timer 0 of AT89C51 in 16-bit mode. A 12MHz crystal oscillator is used for giving a base frequency

Test Points

Test point	Details
TP0	0V
TP1	High when timer becomes zero
TP2	High when switch S4 is opened
TP3	Low when switch S3 is pressed

of 1ms. Timer registers TH0 and TL0 are loaded with values 3CH and B0H, respectively.

The time delay value is calculated by $(65,536 - 50,000) = 15,536 = 3CB0H$, where 50,000 indicates the required time delay in milliseconds and $2^{16} = 65,536$. The interrupt program is used to increment seconds, minutes and hours every one second (50ms $\times$ 20 iterations). The LCD is used to display seconds, minutes and hours after the conversion of values into ASCII characters since LCD controller HD44780 only accepts ASCII characters. When the real-time clock shows the time 12:00:00 seconds, session AM will change to PM, and vice versa. The microcontroller has a reset circuit to reset the clock to 01:00:00.

Construction and testing

An actual-size, single-side PCB for the long delay timer with real-time clock display is shown in Fig. 2 and its component layout in Fig. 3. Assemble